

Assessing Spatial Variability of Soil Nutrients and Topographic Influences for Variable-Rate Fertilization in an Irrigated Corn Field

1.0 Introduction and Research Background

Recently, the transition from conventional uniform field management to Site-Specific Crop Management, also known as precision agriculture, is gaining prominence (Mulla, 2013). Traditional agricultural practices typically treat an entire field as a single unit, applying uniform rates of water, pesticides, and fertilizers, ignoring the significant spatial variability in soil chemical properties, physical characteristics, and micro-topography (Wollenhaupt et al., 1997). Applying a uniform fertilizer rate across a heterogeneous field can result in nutrient-rich zones receiving excessive applications, leading to environmental degradation and unnecessary economic costs, while nutrient-deficit zones are under-fertilized, restricting crop development and penalizing yield (Mulla, 2013).

To implement effective variable-rate application (VRA) strategies, growers must accurately map and analyze the spatial distribution of their fields. Soil grid sampling is needed for collecting localized chemical data, including soil pH, plant-available phosphorus (P), and exchangeable potassium (K) (Wollenhaupt et al., 1997). Because physical soil sampling is labor-intensive and costly, Geographic Information Systems (GIS) utilize spatial interpolation techniques to convert discrete point samples into continuous raster surfaces. Inverse Distance Weighting (IDW) is a widely implemented deterministic interpolation method that estimates values at unsampled locations based on a weighted average of nearby points (Gotway et al., 1996).

While chemical macronutrients are fundamental to plant growth, crop performance is also heavily influenced by physical soil properties and field topography (Kitchen et al., 2005). Apparent Soil Electrical Conductivity (ECa) is a rapid, non-destructive proxy measurement for soil physical parameters, specifically soil texture (Corwin & Plant, 2005). Furthermore, micro-topography, modeled via Digital Elevation Models (DEMs), dictates the gravitational movement of water, influencing localized erosion, drainage, and plant-water availability across a landscape (Kitchen et al., 2005). By integrating interpolated chemical soil surfaces, physical ECa layers, and high-resolution DEMs, researchers can delineate highly functional

“management zones” that capture both chemical and physical limitations to crop growth. This project utilizes advanced GIS analysis within ArcGIS Pro to evaluate spatial soil variability and topographic features across an irrigated corn field.

1.1 Research Objective

The core objective of this study is to evaluate the spatial variability of soil chemical and physical properties within an irrigated corn field to determine if a Variable Rate Application (VRA) fertilizer program is agronomically and economically justified. Specifically, this study aims to: 1) Map soil chemical and corn yield variability; utilize Inverse Distance Weighting (IDW) spatial interpolation to transform point (pH, Phosphorus, Potassium, ECa, and Elevation) and crop yield monitor data into continuous, high-resolution 2D raster layers; 2) Apply University of Georgia (UGA) fertilizer recommendation guidelines for corn to reclassify interpolated soil chemical maps into distinct sub-optimal, optimal, and above-optimal zones; 3) Perform a vector-based spatial overlay (Intersect analysis) to identify critical “prescription polygons” where localized soil nutrient deficiencies directly overlap with below-average historical corn yields; 4) Utilize 3D terrain visualization to drape crop performance maps over a high-resolution DEM, establishing whether micro-topography actively drives corn yield, and 5) Synthesize the economic and agronomic findings to provide the grower with a concrete spatial plan, outlining where to apply targeted variable fertilizer and where to prioritize physical structural remediations (such as drainage or variable irrigation).

2.0 Methodology

2.1 Input

The data sources were from a precision agriculture programming coursework taken previously and represent both proximal soil sensing records, grid soil sample data, topography, and crop yield monitors.

2.1.1 Data Capture and Conversions

- 1. Vector Boundary:** The field boundary was imported as a standard shapefile.
- 2. Gridded Soil Chemistry:** Soil test data containing spatial coordinate points for pH, Phosphorus (P, lb/ac), and Potassium (K, lb/ac) were ingested via a point shapefile.

- 3. GeoJSON to Point Conversion:** The corn yield monitor data, soil apparent electrical conductivity (ECa, 0-30 cm shallow), and elevation data were originally stored in GeoJSON format. To import these files into ArcGIS Pro and work with them, the Geoprocessing tool JSON To Features (Conversion Tools) was used to convert them into point data, which were then exported as .shp files to the project's geodatabase for further analysis.

2.1.2 Data processing (Coordinate Systems, Projections and Transformations)

To prevent spatial misalignment and ensure highly accurate linear distances, the following projection workflow was adopted:

- 1. Define Projection:** Some of the data layers were missing their coordinate system definition metadata (no .prj file). The Geoprocessing tool Define Projection (Data Management Tools) was executed to write the geographic coordinate system metadata, setting it to GCS_WGS_1984 (EPSG 4326) to match its native coordinate format.
- 2. Project Tool (Vector Transformation):** Because spatial measurements in degrees (geographic) distort area calculations, all vector point and boundary features were projected to a local projected coordinate system. The Project tool (Data Management Tools) was run sequentially on each data layer, with the Coordinate System set as NAD 1983 UTM Zone 16N (EPSG 6345), utilizing the transverse Mercator projection with linear units in meters.

2.2 Analysis

Raster Analysis

- 1. Raster Interpolation:** To transform discrete vector point measurements into continuous agricultural surfaces across the entire field, raster interpolation was executed using Inverse Distance Weighted (IDW) Interpolation, with an output cell size of 5 meters by 5 meters. The IDW geoprocessing tool (Spatial Analyst Tools > Interpolation) was chosen because soil nutrients, ECa, and yield are highly spatially dependent variables where local values are heavily influenced by their immediate neighbors. In addition, the boundary shapefile was used to set the processing extent and mask under the Environment tab. This was to ensure that the interpolated values fall strictly within the field boundary.

The interpolated layers produced after this analysis include the Potassium, Phosphorus, pH, yield, elevation, and ECa rasters.

2. **Raster Smoothing:** In order to isolate stable agronomic patterns and reduce noise from the interpolated raster surfaces, particularly for the yield map, elevation, and ECa raster, raster smoothing was performed. The geoprocessing tool Focal Statistics (Spatial Analyst Tools > Neighborhood) was run on yield, elevation, and ECa rasters. A 7 x 7 cell rectangular window was applied, with the statistics type set to majority, generating smoothly smoothed, continuous surfaces for these layers and reducing noise that can hinder clear management decisions.
3. **Raster Reclassification:** Reclassification was done on each of the rasters in order to determine whether variable rate application of nutrients is required or not, and to also understand the spatial variation that exists in the field for the yield, elevation, and ECa layers. Soil pH was classified based on the UGA recommendations for corn ([UGA pH recommendation for corn](#)), while the sufficiency levels for Phosphorus and Potassium were obtained from this UGA document: (<https://aesl.ces.uga.edu/publications/soil/cropsheets.pdf>). To classify these continuous smoothed rasters into decision-making zones, the Reclassify tool (Spatial Analyst Tools > Reclass) was used. Soil pH, potassium, and phosphorus were classified based on the UGA guidelines. The yield raster was classified as Below and Above Average yield after obtaining the mean corn yield (187.5 bu/ac). The ECa layer was classified as Low (0-2), Optimal (2-4), and High (4-6).

Vector Analysis

To generate actionable spatial boundaries that a farmer's tractor can interpret, the raster surfaces were transformed into vector polygons and some overlays were performed to identify decision zones for variable rate management.

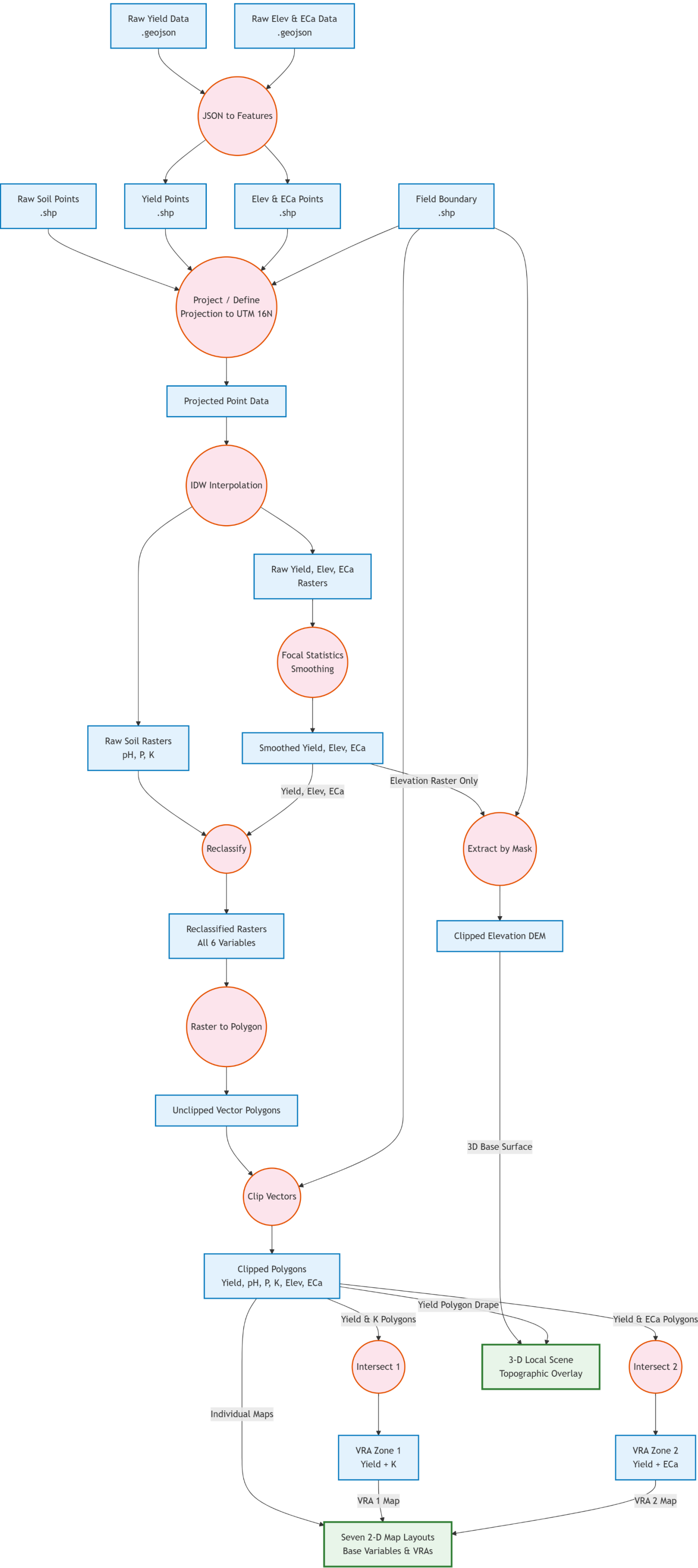
1. **Raster to Polygon Conversion:** All the reclassified rasters were converted to vector layers using the Raster to Polygon tool (Conversion Tools). "Simplify Polygons" was checked to ensure the pixelated 5-meter grid edges were smoothed, turning blocky shapes into smooth, professional, and practical management contours.

2. **Spatial Overlay:** In order to isolate exact areas where soil deficiencies and crop underperformance overlap, the vector Clip tool was run on the polygons using the projected field boundary to produce perfectly bounded vector zones.
3. **The Intersect tool** (Analysis Tools > Overlay) was run with both the potassium and yield clipped polygons selected as input features. This outputted a new combined polygon layer (VRA Zones). The gridcodes for the two layers combined in the new intersected layer were decoded to isolate zones based on the spatial combinations

2.3 Outputs: 2D and 3D Maps

2D maps were created for all important layers, specifically pH, potassium, phosphorus, corn yield map, elevation, ECa, the VRA map showing yield and potassium overlay, and also for yield and ECa overlay. Cartographic elements including a north arrow, scale bar, legend, and data source text were added before exporting. To visually show that the below-average yield band is driven by differences in elevation, a three-dimensional scene was constructed by draping the yield polygons directly over the interpolated digital elevation model. Because the field exhibits minimal natural elevation change, a vertical exaggeration of 20 was applied to the surface to make the central ridge more visible. This 3D view was also exported as a layout with the necessary map elements.

2.4 Flowchart



3.0 Results and Interpretations

3.1 Yield map and Corn Yield Performance

The corn yield map (Figure 1) shows a highly distinct, horizontal band of Below Average Corn Yield (≤ 187.5 bu/ac) running east-to-west through the center of the field. This low-yielding belt is surrounded in the north and south by some Above Average Corn Yield (> 187.5 bu/ac).

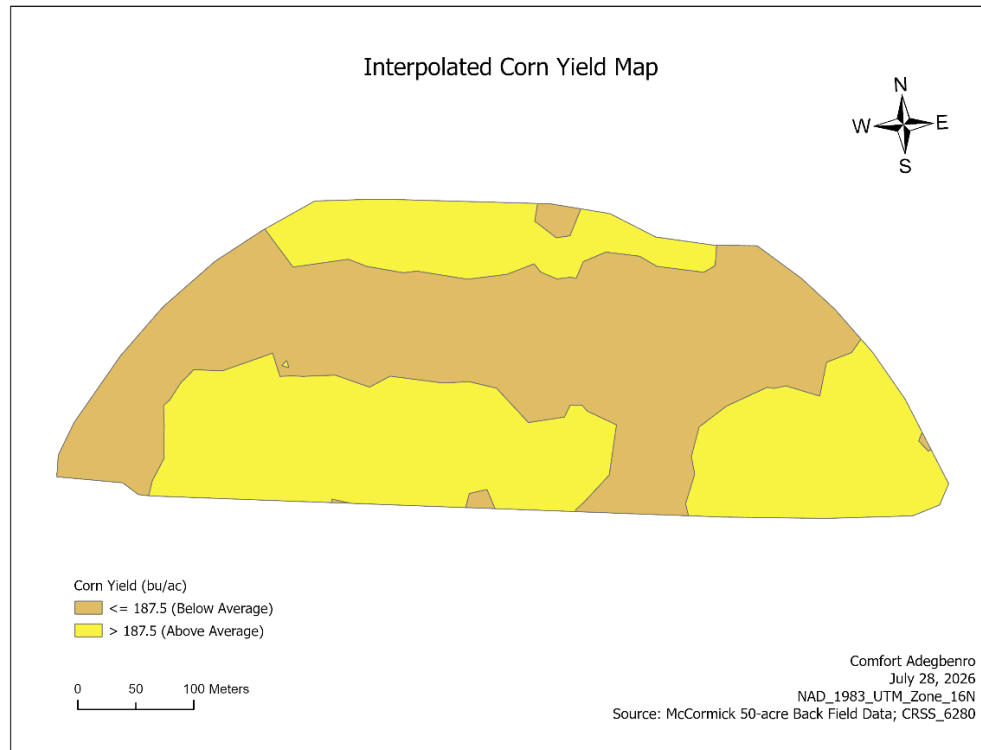


Figure 1: Interpolated corn yield map classified as above and below average yield

3.2 Effects of Soil Chemistry (pH, Phosphorus and Potassium)

To identify the cause for the below-average yield regions, the pH, phosphorus, and potassium maps were explored to map out areas of potential nutrient deficiencies. The interpolated soil pH map (Figure 2) demonstrates that the field is heavily dominated by the Above Optimal (> 6.5) pH category. Only four small, isolated polygons in the central and southern portions of the field fall within the Optimal (6.0–6.5) range. There are also no areas classified as Suboptimal (5.7–6.0) or Inhibitory (≤ 5.7). Hence, soil acidity is completely non-limiting in this field, and no variable-rate lime applications are required.

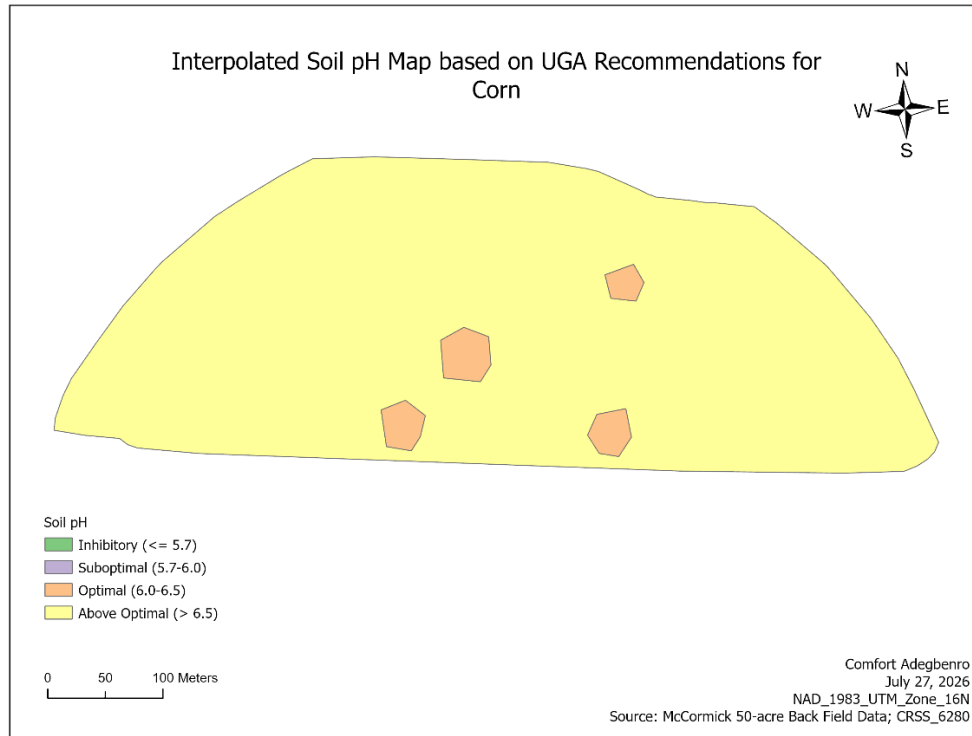


Figure 2: Interpolated soil pH map based on UGA recommendations for corn

The soil Phosphorus map (Figure 3) reveals an abundance of plant-available phosphorus. The majority of the field is classified as Very High (> 100 lb/ac). The remaining portions, located primarily in a large end on the western/southwestern side and another on the eastern/southeastern side, are classified as High (61-100 lb/ac). Importantly, there are zero areas of Low (0-30 lb/ac) or Medium (31-60 lb/ac) phosphorus. Under UGA soil test guidelines, soils in the “High” and “Very High” categories contain more than enough ambient phosphorus to support maximum corn yield. Consequently, there is a 0% probability of a corn yield increase to additional phosphorus fertilizer. From a nutrient-management perspective, Phosphorus application can be completely skipped for this field, resulting in substantial economic savings and eliminating the risk of phosphorus runoff into adjacent water bodies.

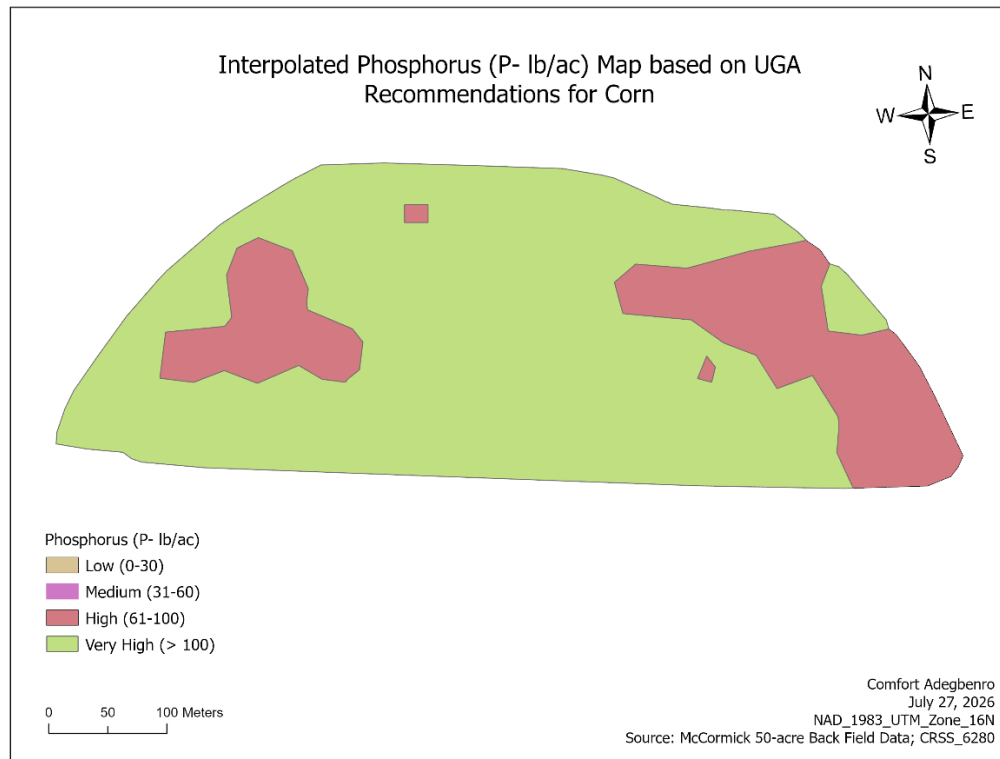


Figure 3: Interpolated soil Phosphorus map based on UGA recommendations for corn

The soil Potassium map (Figure 4) shows a highly sufficient base, but with some exceptions. Approximately 95% of the field area is classified as High (151-250 lb/ac). A single, tiny triangular patch of Very High (> 250 lb/ac) potassium (red) is located on the eastern/southeastern boundary. However, there are four distinct, localized Medium (61-150 lb/ac) level potassium patches: a medium-sized polygon on the western boundary; a larger, blocky polygon near the south-central boundary, and two tiny, isolated triangular/polygonal polygons situated in the center-south. There are no Low (0–60 lb/ac) potassium zones. Under UGA guidelines, soils testing “Medium” for Potassium are considered moderately deficient and have a high probability of showing a positive yield response if potassium fertilizer is applied. In contrast, “High” and “Very High” soils do not require additional inputs. This localized variability represents a classic candidate for a targeted, highly efficient Variable Rate Potassium (K) Prescription. Hence, a variable-rate K application targeting the Medium zones would be cost-effective.

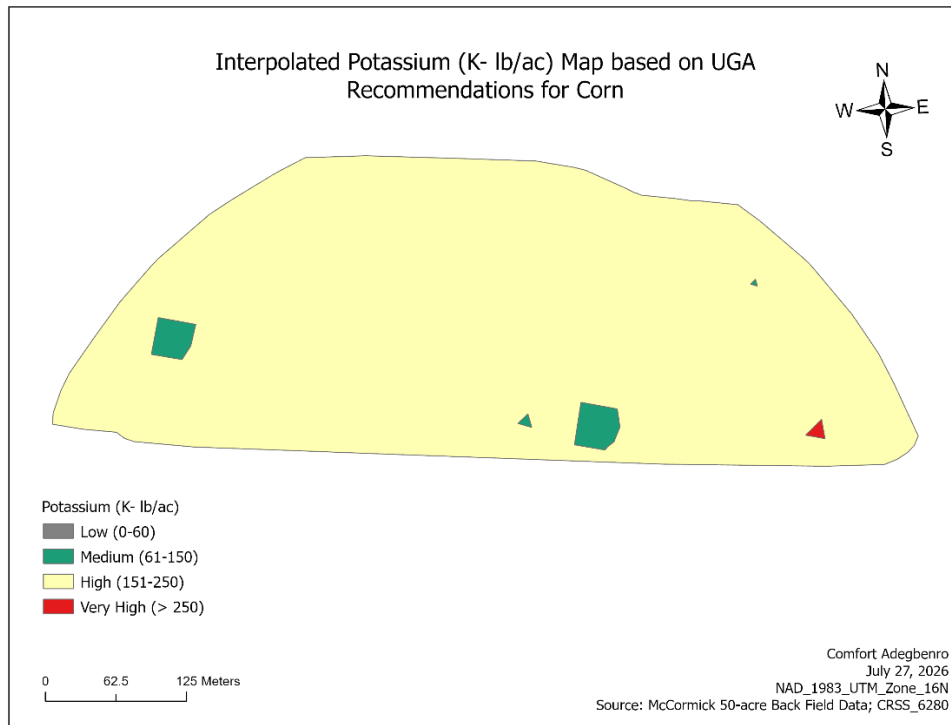


Figure 4: Interpolated soil Potassium map based on UGA recommendations for corn

3.3 Potential Effects of Topography and Soil Physical Properties on Corn Yield

Because soil macronutrients (P and K) and pH were generally non-limiting across the vast majority of the field (Figures 1-3), the extensive spatial patterns observed in the Corn Yield Map (Figure 1) must be driven by other factors, which might be physical, textural, and topographic factors rather than chemical nutrient deficiencies. The crop yield pattern was cross-examined with the shallow soil electrical conductivity (ECa 0-30 cm) and the elevation maps, showing it is heavily dictated by soil physical properties and field terrain, which potentially govern water availability, drainage, and root-zone salinity.

Firstly, the interpolated ECa map (Figure 5) at a shallow depth of 0-30 cm reveals significant spatial variability across the 50-acre study area. The field is characterized by three distinct ECa zones. The Low ECa (0- 2 dS/m) zone covers a substantial portion of the field, across the western, central, and lower-middle sections. The Moderate ECa (2-4 dS/m) category dominates the northern and central-eastern portions of the field. The High ECa (4-6 dS/m) zone is more localized and confined to a distinct region at the northeastern and easternmost margins of the study boundary.

In non-saline conditions, soil ECa acts as a highly reliable proxy for soil texture and water-holding capacity, where lower values correspond to coarser, sandier textures (which have lower clay content and reduced water-holding capacity) and higher values indicate finer clay-loams.

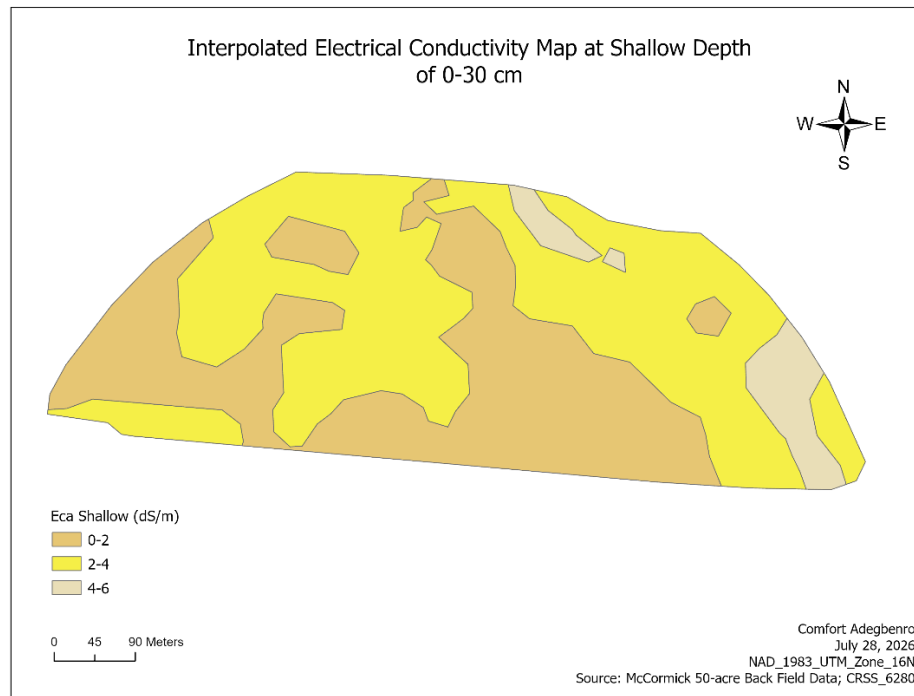


Figure 5: Interpolated soil electrical conductivity (ECa) map at a shallow depth of 0-30 cm

The elevation map (Figure 6) shows a clear elevation gradient across the field, ranging from 50 to 65 meters. There is a high ridge (60-65 m) running diagonally from the center-east to the center-west of the field. Also, the terrain slopes gently downward from this central crown into lower-elevation areas (55-60 m).

3.4 Spatial Alignment of ECa, Elevation and Corn Yield

When comparing these physical layers directly with the interpolated corn yield map, the massive east-to-west horizontal band of below-average corn yield (≤ 187.5 bu/ac) corresponds almost perfectly with the intersection of the high ridge (60-65 m) and the Low ECa (0-2 dS/m) sandy soil. This suggests that this region potentially has coarse, sandy soils with poor water-holding capacity, and when situated on top of a topographic ridge, moisture drains more rapidly away from the root zone. Consequently, the corn crop in this central band potentially suffers from moisture deficits and water stress, which may suppress yields despite the abundant presence of Phosphorus, Potassium, and optimal soil pH. Conversely, the areas of

above-average yield in the northern and southern zones correspond to lower elevations where moisture naturally accumulates, and the soil texture is loamier (Moderate ECa of 2-4 dS/m), establishing an ideal soil-water-nutrient balance.

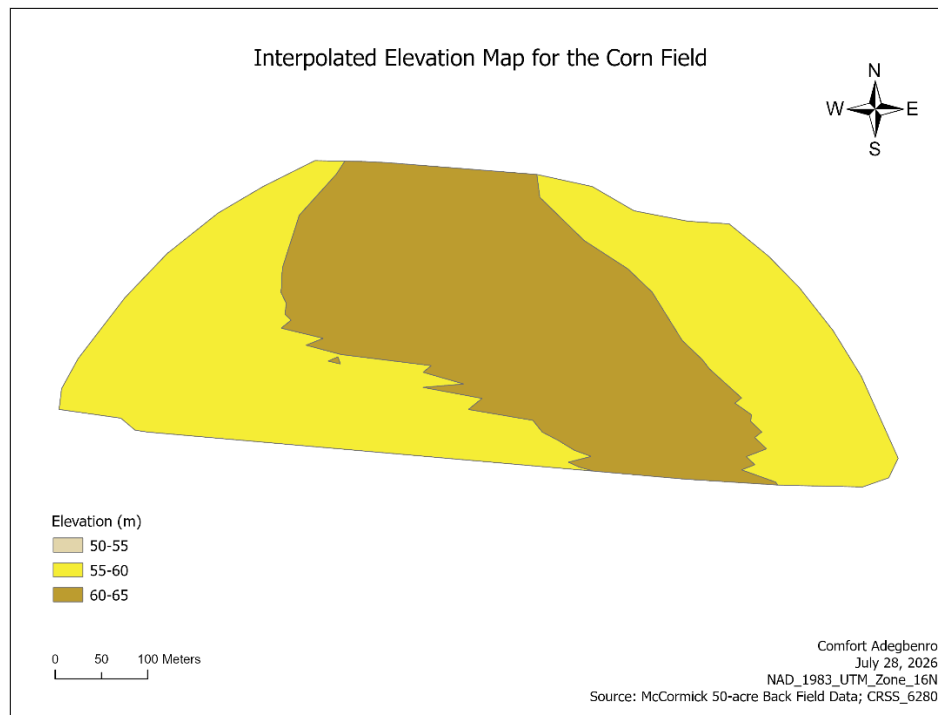


Figure 6: Interpolated elevation map for corn field

3.5 Spatial Overlay Analysis and 3D Visualization for Precision Management Decisions

To translate these spatial correlations into actionable, site-specific management decisions, multiple vector overlay analyses and a 3D topographic scene were developed in ArcGIS Pro.

Specifically, the vector overlay between the reclassified soil Potassium and corn yield layers resulted in five distinct spatial combinations, showing exactly where nutrient deficiencies overlap with crop performance (Figure 7). Below-average yield and medium K is the critical, target Variable Rate Application (VRA) zone, located in a few small, isolated patches in the western and central portions of the field. In these areas, the soil exhibits deficient Potassium levels (61-150 lb/ac) while crop yield is lagging (≤ 187.5 bu/ac). Applying targeted potassium fertilizer only to these polygons is highly agronomically justified to maximize the crop's yield potential. The above-average yield and medium K include the small areas, showing that the crop is currently yielding well despite lower potassium levels, indicating a zone to monitor

closely for future nutrient depletion. Lastly, the High & Very High K zones represent the vast majority of the field. Because soil potassium is already at non-limiting, “High” or “Very High” levels, applying more fertilizer is agronomically useless. By implementing a “zero-application” or maintenance-only VRA strategy across these dominant zones, the grower can skip fertilization entirely, saving substantial input costs and preventing agricultural runoff.

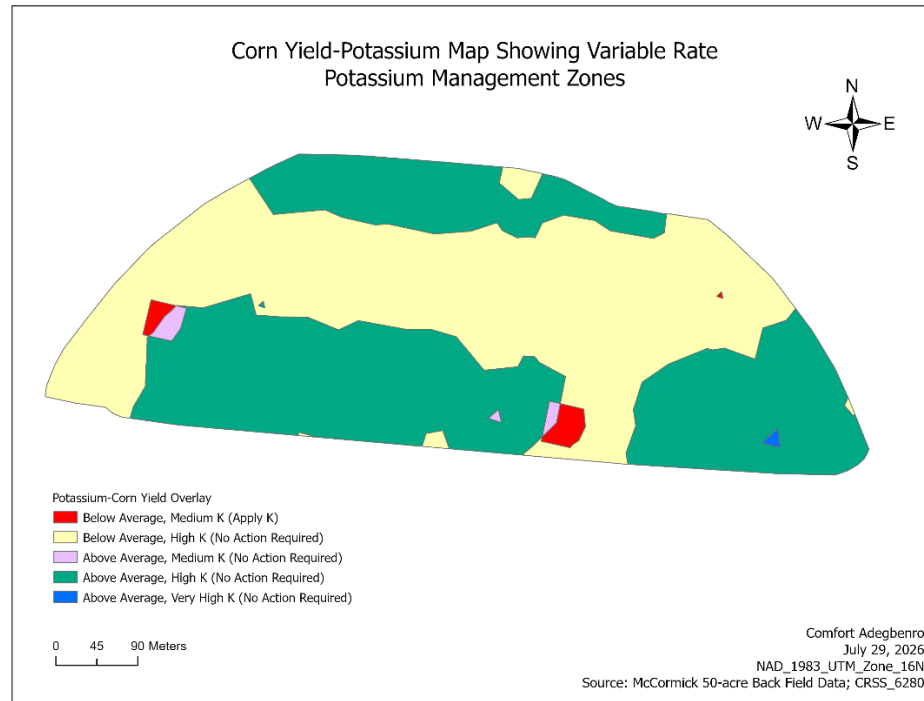


Figure 5: Interpolated soil electrical conductivity (ECa) map at shallow depth of 0-30 cm

While the potassium VRA map provides a chemical prescription, the ECa-Corn Yield Overlay Map provides a physical diagnostic tool by isolating soil texture limitations. Below-average yield, Low ECa cuts through the center of the field, isolating the sandy, drought-prone soils. Here, fertilizing with K or P will not fix the yield gap because the limiting factor is water retention. The recommended VRA strategy is to implement Variable Rate Irrigation (VRI) or split nitrogen applications to prevent rapid leaching through the sandy profile. On the other hand, management practices that can improve water retention can be adopted. The below-average, High ECa zone represents heavy clay in a low-lying basin where soluble salts accumulate, causing osmotic stress. The management recommendation here is to improve subsurface physical drainage to flush out accumulated salts, rather than adding fertilizer. The above-average yield and

optimal ECa zone represent the highly productive, loamy “sweet spots” of the field, where water retention and soil texture are naturally optimal.

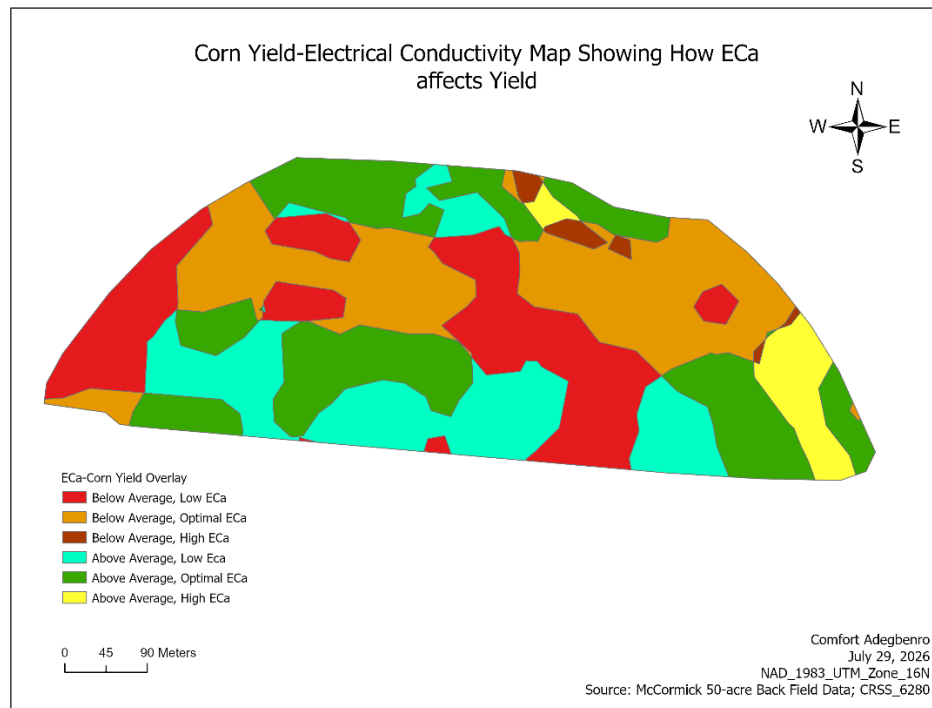


Figure 7: Corn yield and soil electrical conductivity (ECa) overlay map

3D Corn Yield and Elevation Scene

To visually validate the topographic-texture relationship, the reclassified corn yield polygons were draped over the field's continuous elevation DEM. The below-average Yield band sits on top of the raised central ridge, while the above-average yield zones fall into the lower-lying basins. This map illustrates the terrain's influence on crop performance, demonstrating that the primary yield-limiting factor across this field is not a chemical nutrient deficiency, but rather a topographically driven soil moisture deficit.

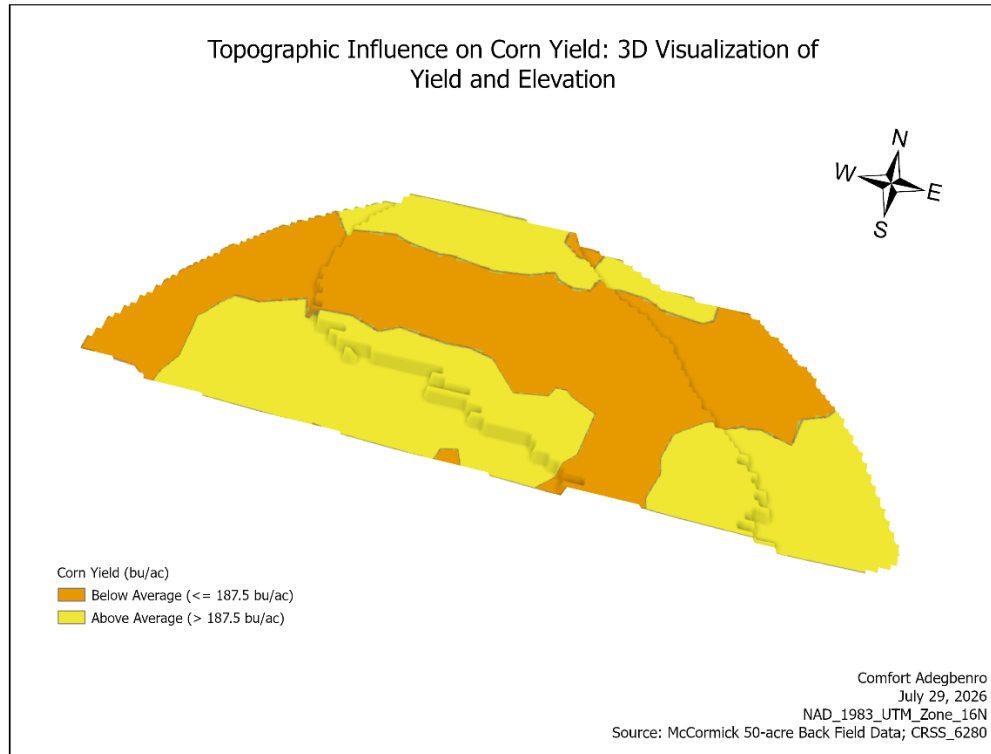


Figure 8: 3D visualization of corn yield and elevation map

Conclusions

This spatial investigation of the field demonstrates that a traditional, uniform application of fertilizer is agronomically and economically unjustified. Applying a flat rate of nutrients ignores the massive spatial variability of the field, leading to severe over-application in highly fertile areas and failing to address localized physical and chemical limitations. By integrating the soil chemistry and physical topography analyses, we recommend a highly profitable, two-pronged management strategy: 1) Implement the Variable Rate Potassium prescription to apply potash only within the “Below Average, Medium K” polygons, with zero rate on the High and Very High zones. 2) Since the central yield depression is driven by topography and soil texture, rather than trying to “fertilize out of the problem,” invest in Variable Rate Irrigation (VRI) for the central dry sandy ridge, and install drainage in the low-lying eastern basin to flush out accumulated salts and restore soil productivity. This strategy maximizes crop yields, reduces fertilizer input costs, and fulfils the agronomic and environmental goals of modern precision agriculture.

References

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